

TAUHUD NABI

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Education

Virginia Tech <i>PhD in Computer Science</i> Advisor: Dr. Shaddi Hasan (Chair - PhD Committee) Committee: Dr. Tijay Chung , Dr. Jeff Reed , Dr. Sang Lee , Dr. Ishtiaque Ahmed (UofT)	GPA: 4.0/4.0	May 2027 (Exp.) <i>Blacksburg, Virginia</i>
Virginia Tech <i>M.S. in Computer Science & Application</i>	GPA: 4.0/4.0 <i>Highest Distinction</i>	Dec. 2025 <i>Blacksburg, Virginia</i>
BRAC University <i>M.Sc. in Computer Science Engineering</i> Advisor: Dr. Golam Rabiul Alam Thesis: <i>Empowering Mobile Network Planning through Deep Learning: A Path to Democratization</i>	GPA: 4.0/4.0 <i>Highest Distinction</i>	Sep. 2021 <i>Dhaka, Bangladesh</i>
American International University - Bangladesh <i>Bachelor in Electrical & Electronic Engineering</i>	GPA: 3.63/4.0 <i>Among top 7%</i>	Feb. 2014 <i>Dhaka, Bangladesh</i>

Research Interests

- Internet Measurement
- AI/ML in Networking
- QoE & Performance Analysis
- Data-Plane Path Inference
- Broadband Mapping & Equity
- Mobile Network Planning

Publications

Under Review

- **Lost in the Prefix: Revisiting IP Geolocation Accuracy Across Networks and Geographies** (Under review, preprint available on arXiv)
Focus: Large-scale evaluation of four major IP geolocation providers across *175 countries* using RIPE Atlas and UNICEF Giga ground truth, showing that *mobile networks* and *Global South* geographies suffer order-of-magnitude higher errors, traced to coarse BGP prefix granularity. [Preprint](#)
Authors: [Syed Tauhidun Nabi](#), Jocelyn Bliton, Tijay Chung, Shaddi Hasan

Peer-Reviewed

- **Red is Sus: Automated Identification of Low Quality Service Availability Claims in the US National Broadband Map** (Full Paper; Accepted and presented in *ACM IMC'2024*)
Focus: Built a *Machine Learning classifier* by combining regulatory and crowdsourced broadband datasets to create a labeled training set for detecting ISP coverage misreporting in the FCC's Broadband Data Collection, spanning the entire *United States*. [Paper link](#)
Authors: [Syed Tauhidun Nabi](#), Zhuowei Wen, Brooke Ritter, Shaddi Hasan
Conference: [ACM Internet Measurement Conference 2024](#)
Location: Madrid, Spain on Nov 4–6, 2024.
- **Deep Learning Based Fusion Model for Multivariate LTE Traffic Forecasting and Optimized Radio Parameter Estimation**
Focus: Designed a deep learning fusion model on 6.2M LTE time-series real world traffic samples to forecast sector-level traffic and resource utilization. Developed algorithms for proactive radio parameter tuning to improve cell performance, enabling data-driven traffic engineering and congestion mitigation in large operator networks. [Paper link](#)
Authors: [Syed Tauhidun Nabi](#), Md Rashidul Islam, Md Golam Rabiul Alam, Giancarlo Fortino
Publisher (Journal): IEEE Access; Date of Publication: Feb 10, 2023.

Research Projects & Presentations

Signal Sense: Measuring Mobile Broadband Inequality | *QoE Metrics, Measurement Blind Spot* Jan' 25 – Present

- *Research Output:* Poster accepted to [ACM SIGCOMM 2026](#); full paper in preparation for *ACM SIGCOMM 2027*.
- *iQoE Metric:* Built *indicative Quality of Experience (iQoE)*, a normalized broadband-quality index from large-scale crowdsourced data (Ookla, M-Lab), exposing disparities obscured by raw KPIs and administrative aggregation.
- *Measurement Invisibility:* Showed measurement density correlates strongly with wealth, rendering marginalized communities statistically invisible (e.g., Korail slum inside affluent Gulshan-Banani); extending the work to passive measurement and a field study.

NetBreakouts: Mapping Mobile Network Egress Topology | *Traceroute, IPv6 to Topology, ML* Apr'25 – Present

- *Research Output:* Early results presented at the [ACM IMC 2025 PRIME Workshop](#); full paper in preparation.
- *IP-Level Inference:* Designed a scalable methodology using traceroute and BGP data to identify and geolocate mobile core gateways (P-GW/UPF) across U.S. operators.
- *IPv6 Structure Analysis:* Decoded hierarchical IPv6 bit patterns to infer regional and subregional router topology, enabling carrier-level segmentation.
- *Data Pipeline:* Built Python/Go pipelines for large-scale traceroute parsing, prefix clustering, and longitudinal tracking of carrier infrastructure.

Carrier WiFi with Magma Core | *Wireless Systems, Edge Offloading*

Jan'20 – Jul'21

- *Deployment:* Implemented carrier WiFi offloading integrated with LTE core using Facebook's Magma Core.
- *Impact:* Enabled heterogeneous spectrum offloading and improved performance in congested regions.

Work Experience

Cisco ThousandEyes

May 2026 – Aug 2026

PhD Intern – Data Science & Research (Internet Measurement)

Remote / San Francisco, CA

- Extended ThousandEyes' global active-measurement event-detection pipeline to make cellular (4G/5G) a first-class component, building a network-layer identity (carrier ASN + H3 geo-cell + radio generation); validated on 51K+ measurements at 82% correct attribution.
- Exposed mobile-core opacity: quantified session-IP churn inflating gateways to ~3 per cellular agent/day vs. ~0.3 for WiFi, masking shared PGW/UPF infrastructure, and scoped path-trace analysis to recover the true gateway map.
- Root-caused a ~55% cellular telemetry gap to an untracked Android permission, driving a fix that raised field availability ~15%.

Virginia Tech – Dept. of Computer Science

Aug 2022 – Present

Graduate Research Assistant (Internet Measurement & Data Systems)

Blacksburg, Virginia

- Led analysis behind an [ACM IMC 2024 paper](#) demonstrating strong predictive performance ($AUC > 0.98$) for identifying low-quality availability claims across 900+ ISPs nationwide, enabling targeted review of ISP self-reported data.
- Developed large-scale Python pipelines to process trillions of traceroute, BGP, and speed-test records (M-Lab, FCC BDC, Ookla) for broadband validation and infrastructure mapping.
- Designed scalable heuristics to infer and geolocate mobile core gateways (P-GW/UPF) and carrier backbone routers using IPv6 structural patterns and prefix clustering.
- Led a large-scale evaluation of four major IP geolocation providers across 175 countries using RIPE Atlas and UNICEF Giga ground truth, tracing elevated mobile and Global South errors to coarse BGP prefix granularity.

Robi Axiata Ltd.

May 2017 – Aug. 2022

Manager – Wireless Systems Engineering & Data Analytics

Dhaka, Bangladesh

- Built large-scale network analytics pipelines using LTE measurement reports and IP-level telemetry to geolocate cell sites and detect routing anomalies with sub-100m precision.
- Designed an internal IP-to-cell mapping framework that reduced drive-test costs by 91% and enabled automated network performance auditing.
- Optimized TCP/IP and RAN parameters through data-driven modeling, improving network throughput and user QoE by 17%.

Banglalink (VEON)

Aug 2014 – Apr 2017

Specialist Engineer – Access Networks & Performance Analytics

Dhaka, Bangladesh

- Monitored and optimized end-to-end cellular network KPIs using real-time traffic and IP telemetry to enhance nationwide infrastructure reliability.
- Implemented automated frequency planning and mobility optimization, improving spectral efficiency across regions.
- Contributed to early validation of next-generation RAN and core designs, focusing on performance, coverage, and IP-level connectivity.

Leadership/ Voluntary

Vice President, CSGC@VT

Aug. 2024 – Jul. 2025

Computer Science Graduate Council

Virginia Tech

- **Facilitated Smooth Operations:** Ensured efficient functioning of the Computer Science Graduate Council, chairing meetings in the President's absence and overseeing various committees, representing around 800 CS graduate students.
- **Championing Student Engagement:** Organized key departmental events and represented the CS graduate student body in various committees, fostering student advocacy and community building.

Honors/ Awards

- **Internet Society Pulse Fellow, 2026:** Awarded the prestigious **ISOC Pulse Fellowship** in Internet Resilience, a highly selective program for early-career researchers
- **ACM SIGCOMM 2026 NSF Travel Grant:** Awarded a competitive travel grant by the U.S. National Science Foundation to attend and present research at ACM SIGCOMM 2026
- **Pratt Fellowship, 2026:** **Departmental Excellence Award** — awarded by the Department of Computer Science (VT)
- **Google Cloud Research Credits, 2026:** Awarded ~\$1,000 in GCP compute credits to support PhD research
- **Third Best Undergraduate Research Poster, 2024:** Supervised undergraduate research team awarded third place at the [2024 Virginia Tech Undergraduate Research in Computer Science competition](#)
- **Best Research Publication Award:** Recognized for outstanding research publication award at BRAC University
- **Highest Distinction/Summa Cum Laude:** Achieved a perfect 4.0/4.0 CGPA in Master of Science in Computer Science and Engineering
- **Best Performance Award:** Honored multiple times by the Technology Division at Robi Axiata during the tenure from 2017 to 2022

Technical Skills

Languages: Python, R, Go, SQL

Data Analytics: Pandas, NumPy, PySpark, BigQuery, Matplotlib, Plotly

Networking & Measurement: Traceroute, BGP, DNS, IPv6 Topology Inference, IP Geolocation, Network Resilience

Machine Learning: Tree-based Models, Deep Learning, Large Language Models, Anomaly Detection, ML Pipelines

Tools & Platforms: Linux, Docker, Git, Jupyter, Google Cloud Platform, AWS (SageMaker, EMR)